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This is a copy of a conversation between ChatGPT & Anonymous.

Report conversation

I am thinking lately about a new research direction which I would like to get your thoughts on.

We discussed in the past about affordance and interfaces, and specifically the setup in which two agents interact as in matting.

We further discussed the example of a stone (or a chair) which affords sitting e.g, for human but not for e.g., an ant.

We further discussed the differences in setups where agent can manipulate its interface (in contrast to e.g., chair) and by that controls its affordance to other agent, as possibly happen in e.g., matting.

Given these, I have been thinking about a system with one sender (or requester/ inquirer) and multiple potential responders.

The responders have information that the inquirer needs, i.e., as in partially observed environment.

Now, the goal of each of the responders is to make the inquirer inquire itself and not one of its peers.

In other words, the reward is not necessarily given for correctly answering the inquirer request but purely for getting engaged, as in matting, where we can assume that once engaged, task is accomplished.

My question thus is what prior art exist for such setup and your insight about it in general

This is a very interesting direction—and it actually sits at a crossroads of several literatures, but doesn't quite match any of them cleanly. That's a good sign: it suggests

you may be carving out something genuinely new rather than just rephrasing an existing paradigm.

Let me try to structure the landscape and then give you a sharper interpretation of what you're proposing.

1. Closest prior paradigms (but none are exact matches)

(a) Signaling games & cheap talk

Classic work (e.g., Michael Spence, Vincent Crawford) studies agents who:

- want to influence a receiver's action
- but may have **misaligned incentives**

Connection to your setup:

- Responders try to influence the inquirer's *choice of whom to query*
- Messages are strategic, not necessarily truthful

Key difference:

- In standard signaling → reward is tied to *outcome correctness*
- In your setup → reward is tied purely to **being selected (attention capture)**

👉 So you are removing *truth pressure* and isolating **engagement pressure**

(b) Rational Speech Acts (RSA)

Work by Noah Goodman models speakers who:

- choose messages to maximize listener inference

Connection:

- Responders optimize messages anticipating the inquirer's reasoning

Difference:

- RSA assumes cooperative goal (correct inference)
 - Your setup is **competitive and self-serving**
-

(c) Attention / advertising auctions

This is surprisingly close.

Think:

- Google ad auctions
- recommender systems (YouTube, TikTok)

Connection:

- Agents compete for **attention**
- Reward = being clicked/selected
- Truth is secondary or even irrelevant

Difference:

- Those systems are *engineered*, not emergent communication systems
- No explicit study of **language emergence under attention competition**

👉 This is probably your *closest real-world analogue*

(d) Multi-agent emergent communication (your home turf)

Your prior work + literature:

- referential games
- Lewis signaling games
- coordination games

Connection:

- Agents develop protocols under task pressure

Difference:

- Typically:
 - success = correct task completion
- In your proposal:
 - success = **being chosen as the communication partner**

👉 This flips the objective from **information transmission** → **partner selection**

(e) Mechanism design / information design

Think:

- Roger Myerson
- Bayesian persuasion

Connection:

- Agents strategically reveal information to influence decisions

Difference:

- Still tied to downstream utility
 - You isolate **pre-interaction competition**
-

2. What is fundamentally new in your formulation

Let me sharpen what I think is the core novelty:

You are proposing a pre-communication game

Instead of:

"What message helps me solve the task?"

You ask:

"What message makes *me* the chosen interface?"

This introduces a new axis:

Classical EC	Your setup
Optimize meaning	Optimize selection
Truth pressure	Weak or no truth pressure
Communication = task solving	Communication = access control
Fixed interface	Strategically shaped interface

3. Conceptual framing (this is where it gets interesting)

I think your idea can be framed as:

Affordance Shaping as Communication

Each responder is not just sending a message, but:

- **designing an interface**
- that changes how the inquirer perceives:
 - usefulness
 - accessibility
 - compatibility

This connects beautifully to your earlier intuition:

Agents can manipulate their interface → thereby control affordances

So instead of:

- chair affords sitting (fixed)

You have:

- responder *chooses how it affords querying*
-

4. What dynamics should we expect?

This is where I think your setup becomes very rich.

(a) Emergence of “hooks” instead of “meanings”

Messages may evolve to:

- grab attention
- signal usefulness
- mimic familiarity

Analogues:

- clickbait
- branding
- style alignment

👉 Language may become:

optimized for **selection**, not **truth**

(b) Specialization vs deception

Two competing equilibria:

1. Honest specialization

- “I am good at X”
- Inquirer learns routing

2. Strategic misrepresentation

- “I claim I’m good at everything”
- leads to:
 - noise
 - adversarial signaling

👉 Question:

What pressures restore honesty (if any)?

(c) Interface convergence

Responders might:

- adapt to match the inquirer's internal representation
- e.g., mimic its embedding space / language style

This is *very aligned* with your "Beyond NL" thinking:

- communication adapts to **receiver affordances**
-

(d) Meta-communication layer

A two-stage protocol may emerge:

1. **Selection message** (get chosen)
2. **Task message** (solve problem)

👉 This separation is rarely studied explicitly

5. Concrete related ML directions (partial overlaps)

You might want to look at:

- "learning to defer" / "mixture of experts routing"
- "query routing" in LLM systems
- "tool selection" in agents
- "active information seeking"

But importantly:

None of these model **strategic competition between responders**

6. A crisp way to position your idea

If I were to phrase your contribution:

We study the emergence of communication protocols under competitive pressure for selection rather than task success, framing communication as affordance shaping rather than information transmission.

Or even sharper:

From meaning to access: communication as competition over who gets to be asked.

7. Concrete experimental setup (quick sketch)

This could be very clean:

- Inquirer sees:
 - task
 - short messages from K responders
- Chooses one responder
- Only that responder answers
- Reward:
 - responders: selected or not
 - optionally: downstream task success

You can then vary:

- alignment between selection and correctness
 - bandwidth constraints
 - observability
-

8. My main insight / suggestion

The strongest conceptual angle (IMO):

👉 **Decouple communication into two functions:**

1. **Access / selection**
2. **Information transmission**

Your setup isolates (1), which is:

- underexplored
 - highly relevant to real systems (LLM agents, APIs, tools)
-

9. A deeper philosophical connection (fits your thesis)

This connects tightly to your "Beyond Natural Language" theme:

- Natural language evolved for:

- shared human affordances
- Your setup asks:
 - what happens when **affordances are strategic and mutable?**

Then:

Language may evolve not to describe the world—but to control interaction topology.

10. Bottom line

- There is **related work**, but no direct match
 - The closest analog is **attention competition systems (ads, recommenders)**
 - The novelty is:
 - shifting from *truthful communication* → *competitive selection*
 - framing communication as **affordance engineering**
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If you want, next step we can:

- formalize this as a game (utility, equilibrium)
- or design a minimal experiment (very much like your referential setups but with routing)

Few things.

First, I used the matting scenario as an example while you did not refer to it at all. Is matting the correct word? Do I spell it correctly?

Second, i assume multiple turns/interactions are needed for balancing capability (honestly, task success) with attractiveness (advertising), Don't you think.

And third, I agree that the ad/recommender is the closets one and wonder if it not already covers my idea entirely

Great—these are exactly the right pressure points to clarify. Let me go one by one.

1. "Matting" — is this the right term?

Short answer: **no**—this is almost certainly not the word you want.

- *Matting* is a term from image processing (alpha matting, foreground extraction).
- It has no meaning in interaction/game/communication contexts.

What you probably meant

From your description ("agents trying to get engaged"), the closest terms are:

- **Mating** (evolutionary biology)
- **Courtship signaling**
- **Mate selection**

These are *very* well-aligned with your idea:

- Agents compete to be **chosen**
- Signals evolve to maximize **selection probability**
- Truthfulness is secondary or even irrelevant

This connects directly to signaling theory (e.g., Amotz Zahavi and the handicap principle).

👉 So yes—you likely meant "**mating**", not "matting".

And actually, this analogy is *extremely powerful* for your setup.

2. Do you need multiple turns to balance attractiveness vs capability?

Short answer: **yes**—and this is a key insight, not just an implementation detail.

Let me sharpen it:

Single-shot setting → degeneracy

If selection is based only on a single message:

- Optimal strategy = **maximize attractiveness**
- No pressure for:
 - truth
 - competence
 - reliability

→ You get:

- clickbait
- deception

- collapse of informativeness
-

Multi-turn interaction → introduces accountability

Once you allow:

1. Selection phase
2. Interaction phase
3. Outcome feedback

you create a tension:

- Short-term reward: get selected
 - Long-term reward: be selected *again*
-

This leads to a fundamental tradeoff

You get something like:

Attractiveness vs Credibility

Which mirrors:

- advertising vs product quality
 - signaling vs substance
 - your framing: **affordance shaping vs actual affordance**
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Important design knobs

You can control this tradeoff via:

- repeated interactions (same inquirer vs population)
 - memory (does the inquirer remember past success?)
 - delayed rewards
 - reputation mechanisms
-

My take

This is not just “nice to have”—it’s **structurally necessary** if you want:

- non-trivial equilibria
- emergence of meaningful signaling

Otherwise, the system will likely converge to:

"optimal persuasion under zero accountability"

3. Does ads / recommender systems already cover your idea?

This is the most subtle question—and the answer is:

They are very close functionally, but they do *not* cover your idea conceptually.

Let me explain carefully.

Where they *do* overlap

Ad systems (e.g., Google, Meta Platforms):

- Agents compete for **selection (clicks)**
- Signals (ads) are optimized for **engagement**
- There is a separation:
 - ad → selection
 - product → outcome

This is almost exactly your two-stage idea.

But here's the critical difference

1. No emergent communication

Ads:

- signals are **engineered**
- language is fixed (natural language, images)

Your setup:

- signals can **emerge**
- protocols can **drift away from natural language**

👉 This is a *huge* difference (and very aligned with your thesis)

2. No strategic co-adaptation of protocols

Ads:

- optimization is mostly:
 - individual (per advertiser)
 - against a fixed platform/user model

Your setup:

- responders + inquirer can **co-evolve**
 - communication becomes:
 - adaptive
 - potentially compositional
 - possibly non-human-readable
-

3. No explicit modeling of "affordance shaping"

Ads optimize:

- click-through rate

But they don't model:

"How do I change the interface I present so that I become the preferred communication partner?"

That is much closer to:

- *protocol design*
 - *representation alignment*
-

4. No clean separation of the scientific question

Ad systems conflate:

- selection
- task success
- user satisfaction
- platform incentives

Your formulation isolates:

selection as the primary objective

That isolation is scientifically valuable.

So how should you think about it?

I'd phrase it like this:

Ad systems are an *instance* of your phenomenon under fixed human language and engineered constraints.

While your work would be:

A general theory of selection-driven communication under adaptive protocols.

4. A deeper unifying perspective (this is the exciting part)

If we connect all three points:

- mating (selection pressure)
- multi-turn interaction (reputation)
- ads (attention economy)

You get:

Communication under selection pressure

This is different from:

- communication under cooperation (RSA)
- communication under coordination (referential games)

Instead:

Agents communicate to **control who gets to communicate further**

5. A concrete conceptual upgrade (suggestion)

I think your strongest framing is:

From referential games to routing games

Where:

- the core decision is:
 - *who to talk to*, not just *what to say*

And signals are:

not descriptions of the world, but **proposals for interaction**

6. Bottom line answers

- **"Matting"** → incorrect
→ you likely mean **mating / courtship signaling**, which is actually a *great* analogy
 - **Multiple turns** → not just helpful, but likely **essential** for meaningful dynamics
 - **Ads/recommenders** → closest real-world analogue, but:
 - they **do not capture emergent protocol formation**
 - they **don't isolate the scientific question**
 - they **lack co-evolving communication**
-

If you want, next step I'd suggest is:

- write a **minimal formal game** (players, messages, selection rule, reward)
- and check what trivial equilibria exist (this will clarify what constraints you *must* add)

Here is an initial idea based on the referential game setup I used in my last paper available in arxiv.

Assume a sender and multiple receivers, maybe each based on a different LLM.

Now assume 4-turn game. 3 warm-ups and the last one with high reward.

In each of the three first turns, the sender describes a target out of n candidates. In these turns the receivers know the target.

Now, the receivers can choose how to answer. They can either simply provide the sender with the answer, or they can explain how they conclude that this is the answer.

Then, in the fourth turn that receives do not know the target and the sender needs to choose to which of them it wants to address its last description.

What do you think? Please be critical here and suggest improvements as you find fit

Thought for a couple of seconds

I think this is promising, but in its current form it still has a few conceptual leaks. The good news is that they are fixable, and fixing them will likely sharpen the paper a lot.

The strongest part of the setup is that it tries to operationalize exactly the tension you care about:

receivers are not only solving the task, but also shaping the sender's future choice of partner.

That said, here is the critical view.

What already works well

The 3 warm-up + 1 high-stakes turn structure is a good instinct because it introduces:

- a **history**
- a chance for receivers to build **reputation**
- a clean separation between **demonstration** and **selection**

This is much better than a one-shot setup, because one-shot would mostly collapse into "say whatever makes me attractive."

Also, making the final turn the only one where receivers do not know the target is clever, because it forces the sender to use earlier interactions to infer which receiver is worth querying.

So the core idea is sound.

Main concern 1: the receiver's action space is too weak

Right now the receiver can:

- give the answer
- explain the answer

This is a nice first pass, but I think it is too narrow. Why?

Because "explaining" may just become:

- longer answer = better
- more natural sounding answer = better
- stylistically persuasive answer = better

Then you are partly measuring verbal fluency or style compatibility, rather than the thing you actually want:

how receivers shape their interface to become selected.

I would broaden the receiver action space. For example, let the receiver choose among:

- answer only
- answer + explanation
- answer + confidence
- answer + short rationale
- answer + promise of future usefulness
- answer in a receiver-specific protocol or format

Even better, make the output budget fixed, so they must decide how to spend it:

- all budget on direct answer
- some on rationale
- some on self-presentation
- some on calibration

That would create a much more meaningful tradeoff.

Main concern 2: “receivers know the target” may make the warm-up too artificial

This is the point I would push on hardest.

If the receivers know the target in the first three rounds, then those rounds are not really about inference. They are almost like staged auditions. That is not necessarily bad, but then the interpretation changes.

You are no longer studying:

- “which receiver is actually better at the task?”

You are studying:

- “which receiver performs best when already privileged with ground truth?”

That risks making the warm-up phase more like a demo script than an epistemic interaction.

I think you should seriously consider one of these alternatives:

Better option A

In the first three rounds, receivers do **not** get the target, but they do get the same candidate set and sender description, and must infer the target.

Then the sender observes:

- correctness
- style
- explanation quality
- calibration

This would make the warm-up rounds a true measure of competence plus interface quality.

Better option B

If you want privileged information in the warm-up, then make that privilege partial and asymmetric:

- each receiver sees only some extra feature
- or a different view of the target
- or a compressed latent hint

Then the sender learns not only “who sounds good” but “who is good for which kind of information.”

That would be richer and more aligned with your partially observed environment motivation.

Main concern 3: the final round may still collapse into “pick the one that sounded nicest”

In the current setup, unless you design incentives carefully, the sender’s final choice may mostly reflect:

- linguistic similarity
- verbosity
- apparent confidence
- perceived friendliness

In other words, partner selection may become a style preference task.

That is not worthless, but it may be less novel than you want.

You need the sender’s choice to be grounded in something more robust, such as:

- past accuracy
- calibration
- specialization
- usefulness under certain target types

So I would add task structure that supports receiver differentiation.

For example:

- Receiver A is better on color-based distinctions
- Receiver B is better on spatial relations
- Receiver C is better on fine-grained object identity

Then the sender can rationally choose whom to query in the final round depending on the current target/candidate set.

That would let you study **routing based on inferred affordances**, which is stronger than simple preference learning.

Main concern 4: the reward design is underspecified

This is crucial.

Who gets reward for what?

At the moment I infer something like:

- warm-up turns: maybe low reward
- final turn: high reward for selected receiver if task succeeds?

But your earlier framing emphasized that responders may care primarily about being selected, not necessarily about correctness.

You need to decide which world you want:

World 1: pure attractiveness

Receiver reward in final turn depends mainly on being selected.

This will likely produce:

- persuasion
- signaling
- maybe deception

Interesting, but dangerous because it can degenerate.

World 2: attractiveness + competence

Receiver reward depends on:

- being selected
- and task succeeding

This is probably the better first paper, because it avoids total collapse into advertising.

World 3: mixed incentives

Different receivers get different utility weights:

- some care more about selection
- some care more about correctness

This is probably the most interesting scientifically, but maybe too much for a first version.

My suggestion: start with World 2.

Main concern 5: three warm-up turns may be too few

Three is elegant, but I am not sure it is enough for stable partner modeling, especially with stochastic LLM behavior.

With only three rounds:

- a lucky or flashy receiver may dominate
- sender may overfit to shallow impressions

I would treat the number of warm-up rounds as an experimental variable:

- 1
- 3
- 5
- 10

That can reveal whether sender choices become more competence-based with more evidence.

This could become an important result on its own.

Main concern 6: if receivers are fixed LLMs, are you studying emergence or selection?

This is a framing issue.

If each receiver is a different off-the-shelf LLM, and there is no adaptation over time, then the main phenomenon is:

- sender learning whom to trust or query

That is already interesting, but it is not really emergent communication.

It is more like:

- partner selection
- trust formation
- interface preference
- routing under uncertainty

If you want stronger emergence claims, you need at least one of:

- receivers adapt their response style over repeated games

- sender adapts its descriptions to particular receivers
- a protocol emerges between sender and a selected receiver

Otherwise, I would avoid calling it emergence too strongly.

Improvements I would suggest

1. Remove ground-truth privilege in the warm-up

Have receivers infer from the sender's description and candidate set.

This makes early rounds informative about actual capability.

2. Give receivers constrained output budgets

For example, 20 tokens total.

Then they must decide how to allocate capacity between:

- answer
- reasoning
- calibration
- self-presentation

This is where the real tradeoff appears.

3. Create receiver specialization

Deliberately design target sets where different receivers or prompting styles excel on different dimensions.

Then final-round selection becomes a meaningful routing decision.

4. Separate observed variables

For each warm-up round, record:

- accuracy
- answer latency if relevant
- confidence
- explanation length
- explanation consistency
- sender's posterior preference

This will help you distinguish "selected because good" from "selected because appealing."

5. Add a sender-choice baseline

Compare:

- sender picks receiver after warm-up
vs
- oracle picks best receiver
vs
- random receiver
vs
- always same receiver

This will tell you whether the warm-up interaction actually enables useful routing.

6. Consider hidden calibration tests

In some warm-up rounds, include adversarially hard cases where overconfident receivers get exposed.

This could be important if you want to study attractiveness vs reliability.

A stronger formulation

I think a cleaner version would be:

There are one sender and K receivers.

Across T warm-up rounds, the sender describes a target among N candidates.

Each receiver sees the description and candidate set and must identify the target, optionally allocating a fixed message budget among answer, rationale, and confidence.

The sender observes the responses and, after the warm-up, faces a high-stakes round in which it can query only one receiver.

The receivers' utility depends on both being selected and helping solve the final round.

That is already quite crisp.

Even stronger variant

An even better version, in my view, is to make the final round slightly asymmetric:

- sender sees target
- receivers see candidate set
- sender may send a message only to the chosen receiver
- chosen receiver answers
- success depends on joint coordination

Then partner choice matters not just because of "who is smart," but because of **who can coordinate well with this sender**.

That would connect more directly to your earlier affordance/interface line.

My bottom-line judgment

The idea is good.

The core intuition is real.

But the current version is still too close to an audition with stylistic explanations.

The biggest improvements would be:

- remove receiver access to the target in warm-up
- constrain response budget
- make receiver specialization explicit
- reward both selection and correctness
- let adaptation occur over repeated games if you want emergence claims

With those changes, I think the setup becomes much more publishable and conceptually sharper.

The single most important question to settle is this:

Do you want to study persuasive self-presentation, competent routing, or emergent pair-specific coordination?

Right now the setup mixes all three a bit. The paper will get much stronger once one of them is declared primary.

I can help you turn this into a formal game definition next.